NetManager (net_mgr) Training

NetManager is a command line tool used to perform common operations necessary for maintaining NAN devices. The supposed use cases are:

- Access Point (AP) Provisioning/Configuration
- Uploading relay and meter configuration files (BLOBs) to APs
- Collecting device information and statistics from a NIC
- Viewing information about certificates installed on a NIC
- Restarting a NIC
- Managing and setting time on a NIC
- Pushing Certificate Revocation Lists (CRLs) to a NIC
- How to check the clear text port status of a device
- Potential Errors and Troubleshooting steps

Super User cases:

- Closing the clear text port on a device
- Opening the clear text port on a device

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Using NetManager

NetManager consists of a single executable binary file (net_mgr) that is licensed by Silver Spring Networks. It is suggested to place it in a directory included in the shell path. It does not require any special user permission (ie root) to execute.

Usage Syntax:

For most prescribed use cases, the syntax is:

net_mgr -d <NIC's_IPv6_address> <command>

To obtain in-line help for specific commands:

net_mgr –h <command>

A commonly used optional flag is *–t<number>*
This specifies the timeout (in seconds) before net_mgr returns a timeout error

Running net_mgr with no arguments produces the following in-line help:

Usage: net_mgr [-h] [-o ofile] [-s] [-S <v6>] -d <v6>|-i <rf_itf> type [clear]
-h help (this message)
-d address of the ncc you are messaging
-i RF card you are messaging
-t timeout in seconds to wait for a response
-s display OSTime and system_time
-o The file to write the net_mgr request to instead of sending it
-S Set source address in IPv6 message

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Common NetManager Use Cases

The following section identifies the net_mgr commands and arguments to be used with the syntax above, their effect, and their output.

AP Provisioning – Setting and viewing configuration settings

The process of provisioning a new AP includes setting and verifying several required values.

- IPv6 address of DNS server (aka Registrar)
    - View current DNS server
        - Command: *dns server_get*
        - Sample output:
          DNS Server Address: fdbc:78e:c408:1df7:222:19ff:fec7:a805
    - Set DNS server
        - Command: *dns server_set <IPv6 of Registrar host>*
    - View currently configured DNS zone
        - *Command: dns zone_get*
        - Sample output:
          DNS Zone is: prod01.bge.ssnsgs.net
    - Set DNS zone
        - Command: *dns zone_set <DNS_zone_name>*

- IPv6 address of trap server (aka TMB host)
    - This is typically the same IPv6 address of the DNS/Reg server
    - View current trap server
        - Command: *nm_trap host_get*
        - Sample output:
          Trap Host Address: fdbc:78e:c408:1df7:222:19ff:fec7:a805
    - Set trap server
        - Command: *nm_trap host_set <IPv6 of TMB host>*

- IPv6 address of NTP server
    - This is typically the same IPv6 address of the DNS/Reg server
    - View current NTP server
        - Command: *ntp server_get*
        - Sample output:
          NTP Server Address: fdbc:78e:c408:1df7:222:19ff:fec7:a805
    - Set NTP server
        - Command: *ntp server_set <IPv6 of NTP host>*

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- NTP Ops mode
  - Specifies NTP operation mode (should always be enabled, unicast)
  - View current NTP opsmode setting
    - Command: *ntp opsmode_get*
    - Sample output:
      sntp client is ENABLED and operating in unicast mode
  - Set NTP Ops mode
    - Command: *ntp opsmode_set enable*

- Time Zone
  - View current time zone
    - Command: *time local*
    - Verification: Compare the offset (sec) and DST settings to customer's specified timezone
    - Sample output:
      Time Zone info used
      hr:min:sec 13:51:49
      Day of month 11
      Month (start at 1) 12
      Year 2012
      Day of week (sun == 0) 2
      DST on 0
      DST off (0, gmt-changes) 0
      Time zone offset (sec) -28800
  - Set time zone
    - Command: *time_zone set <time_zone>*
    - On Linux, refer to */usr/share/zoneinfo* for valid time zone names

- Network ID
  - Unique numeric identifier associated with each environment
  - **Changes must be followed by a NIC *restart* to take effect!**
  - View current network ID
    - Command: *conf mlme mlme_mac_net_id*
    - Sample output:
      MAC Network ID is set to 100
  - Set network ID
    - Command: *conf mlme mlme_mac_net_id <net_id> (decimal value)*

- Promiscuous Mode
  - Enables or disables the ability for this device to establish a connection to another device on the network that is configured with the promiscuous network ID (65535)
  - **Changes must be followed by a NIC restart to take effect!**
  - View current promiscuous mode setting
    - Command: *conf mlme mlme_ignore_prom_net_id*
    - Sample output:
      MLME ignore promiscuous network ID is set to 0
  - Disable promiscuous mode (i.e. do not allow connections from devices with promiscuous ID)
    - Command: *conf mlme mlme_ignore_prom_net_id 1*

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- Routing
  - Enables or disables the ability for this device to perform routing of NAN traffic
  - Must be enabled for APs to route traffic from associated relays and meters to the application
  - **Changes must be followed by a NIC *restart* to take effect!**
  - View routing status
    - **Note:** Status shown will reflect changes not yet committed
    - Command: *conf srt on*
    - Sample output:
      SRT On is set to 1
  - Verify absolute routing status
    - To confirm routing status at the immediate time:
      - Command: *sysstat*
      - The presence of a running *srt* task (see below) in the output indicates routing is on right now.
      - Sample Output:

        Task List:
        […]
        srt (prio: 10, csw: 116644, stk: 1332/ 8192 16%)

      - The output of *conf srt on* indicates what the routing status will be following a restart
  - Enable routing
    - Command: *conf srt on 1*

- Configuration BLOBs
  - Determine if an Access Point is configured with Secure or Unsecure meter config BLOBs
  - Config BLOBS are stored on the AP in the following sysvars:

AP Config Blob	Non-Secure sysvar ID	Secure sysvar ID
relay (router)	131	554
meter	132	555
bridge	279	556

  - View whether AP has a secure meter config blob defined:
    - Command: sysvar *555*
    - Sample Ouput:
      Sysvar #555 is not defined
  - View whether AP has a secure relay config blob defined:
    - Command: sysvar *554*
    - Sample Ouput:
      Sysvar #554 is not defined

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Other Use Cases: Miscellaneous Operations

- **Restarting a Network Interface Card (NIC)**
  - Instructs a NAN device to reboot its Silver Spring Networks NIC (**not** the meter itself, if run on a meter device)
  - Schedule restart in <n> minutes:
    - Command: *restart in <n>*
  - Restart device NIC immediately
    - Command: *restart now*

- **Managing and setting time on a NIC**
  - Set time to current UNIX time, plus offset of <n> seconds (use *0 for no offset)*
    - Command: *time set_time <n>*
  - *Force AP to sync time via ntp to specified NTP server ipv6 address (see AP configuration section)*
    - *Command: ntp force_req*
  - *Force non-AP device to sync time with associated AP, or optional neighboring device's MAC*
    - *Command: time force <MAC address>*
  - *Enable/Disable time sync on a device*
    - *Command: conf timesync enable [0|1]*

- **Replacing Manufacturing certs**
  - Command: *certs uload_blob <MFGcertblob>*

- **Upload operator or other cert to the NIC**
  - Command: *certs uload <op x509 cert> persist*

- **Pushing CRLs to a device**
  - CRLs can be obtained using Silver Spring Networks CA Tools, and will be provided by Silver Spring Networks
  - Command: *crls uload*

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How to manually check clear text port status on a device

Ensure the device is reachable using ping6; ICMP traffic is still allowed even with the cleartext port off. If the ping fails, LPO cannot be verified; wait for the meter to be reachable.

If the device responds to ping, check the LPO status directly by running the following command:

./net_mgr_lite -d <ipv6> conf nm_sec disable_unsecure

If the command succeeds and returns "0", then the cleartext port is still on.

If the command times out, and the device is pingable, then the legacy port is likely turned off. To be completely sure, use secure net_mgr (net_mgrS) with the appropiate keystore to get the status of nm_sec disable_unsecure.

Example:

```
$ net_mgrS --nm=<path to net_mgr_lite> -d fd79:7414:9ad1:810:213:5001:f4:c3e  conf  nm_sec disable_unsecure
unsecure net_mgr disabled is set to 1
```

```
$ net_mgrS --nm=<path to net_mgr_lite> -d fd79:7414:9ad1:810:213:5001:f4:c3e echo 1
Ok
req id 1844
assoc id 41023
nsar reply type 0x04
Status: OK
inception time: 1335343620, Wed Apr 25 08:47:00 2012
 expiration time 1339148520, Fri Jun  8 09:42:00 2012


Key signer:
0000: 04 08 00 13 50 01 00 f4 0c 3e                     ....P....>
Shared secret is
deb108bbad0a07058b4c17b7a14b44dbdd8e34486f4c9dc
INFO  net_mgrS.c:808   Association to be good for 17 days
Ok
req id 41023
assoc id 41023
nsar reply type 0x00
Status: OK
INFO  net_mgrS.c:793   ASSOCIATION SUCCESS
Ok
Echo result (resp len = 1) :
1
$
```


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Using NetManager to Obtain Device Information and Statistics

In addition to actively configuring devices, NetManager can be used to request a wide variety of information on various aspects of a device's operational performance metrics or other configuration values.

View Certificate Information:

- Displays a list of certificates owned by the NIC

  Command: *certs own*

  Sample output:

```
Certificates owned:
0xc7<BirthCertificate,verifiedBC,ManufacturingCertificate,OperatorCertificate,PrivateKeyOK>
```

- Displays summary of cached certificate information

  Command: *certs sdump 4*

  Sample output:

```
************* Cert Cache **************
Pid:    1024   OP (522 bytes): 2015-06-14 CN=Operator Root,OU=SmartMeter,O=Bobs Electric,C=US
Pid:0x200000  DYN (546 bytes): 2013-03-14 CN=ECBOCA_RD,OU=SmartMeter,O=Bobs Electric,C=US
Pid:0x200001  DYN (566 bytes): 2013-03-14 CN=NM1234_RD,OU=SmartMeter,O=Bobs Electric,C=US
Pid:      17   BC (401 bytes): 2040-02-01 SSNMAC=0013:5001:0140:1621
Pid:      16  MFG (475 bytes): 2039-02-26 CN=S017,OU=MFG,O=SSN,C=US
Pid:       2 root (463 bytes): 2047-08-31 CN=SSN Root Certificate,O=Silver SpringNetworks,C=US
Pid:       1 root (463 bytes): 2047-07-12 CN=SSN Root Certificate,O=Silver Spring Networks,C=US
**************************************
```

View general system statistics:

- Display an overview of device metrics and performance

  Command: *sysstat*

  Sample output:

  System Info

```
I210DT meter (device type: 0x9)
Hardware version: 0x03 revision: 0x00
Firmware version: 2.16.3c

0 CPU Usage
21 Tasks Created
806320785 Context Switches
2145998 Seconds Clock
MBUF size=1788 free=114
treck heap size=18112 bytes free=47424 bytes
heap: beg=03197160 brkval=0335d798 end=33fffd0 total=2526832  left=665656
Uptime:  24 days,  19:06:38 booted at Tue Feb 12 03:04:41 2013

Task List:
            fh (prio:  4, csw: 107995106, stk:   252/ 2048 12%)
           mac (prio:  5, csw:    492375, stk:   232/ 2048 11%)
     wpand_drv (prio:  6, csw:   3890665, stk:   856/ 1024 83%)
         treck (prio:  7, csw:  64306116, stk:   728/ 8192  8%)
          uart (prio:  8, csw: 308269191, stk:   612/ 8192  7%)
```


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```
                 net (prio:  9, csw:  33606805, stk:   640/ 8192  7%)
                 srt (prio: 10, csw:    116732, stk:  1332/ 8192 16%)
                mlme (prio: 13, csw:   6742027, stk:   680/ 2048 33%)
             imu_app (prio: 14, csw:       312, stk:   292/ 8192  3%)
            meter_dl (prio: 15, csw:  27605158, stk:  1932/ 8192 23%)
             net_mgr (prio: 17, csw:    571302, stk:  3288/10240 32%)
             img_mgr (prio: 18, csw:         5, stk:   496/ 8192  6%)
           time_sync (prio: 19, csw:       612, stk:   240/ 4096  5%)
                 led (prio: 21, csw:   4351430, stk:   120/ 1024 11%)
        c1219_tsync (prio: 25, csw:         1, stk:   112/ 2048  5%)
         ZbTickMgrD (prio: 26, csw:  25475261, stk:  1884/ 4096 45%)
                ESP (prio: 28, csw:   2235667, stk:  1152/ 4096 28%)
             crypto (prio: 29, csw:     40404, stk:  3416/12288 27%)
             sanity (prio: 30, csw:    429180, stk:  1164/ 2048 56%)
               stat (prio: 31, csw:   2374098, stk:   120/  512 23%)
               idle (prio: 32, csw: 217818338, stk:    68/  512 13%)
```

Display currently running and stored firmware images:

- Command: *image list*

  Sample output:

```
Last Known Good Image: 02.16.003c
Currently Running Image (flash): 02.16.003c
Currently Running Image (mem): 02.16.003c
Image List:
Image #0: 02.16.003c (crc = f4f2, valid, size = 1260512)
Image #1: 00.00.0000 (crc = 0, empty, size = 0)
```

Viewing other device Statistics:

- Display device NIC's uptime information

  Command: *node_age*

  Sample output:

```
Node age is 2146582 seconds
Node has been up since Tue Feb 12 03:04:41 2013
All times are local to the computer
```

- Display cumulative NIC reboot counter

  Command: conf mlme mlme_reboot_cntr

  Sample output:

```
MLME reboot counter is set to 37
```


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- Display WAN dialer (cellular modem) statistics (AP only)

  Command: *stat wan_dialer*

  Sample output:

```
WAN Dialer Statistics:
   258 Total number of dial attempts
     0 Current number of dial attempts
    62 Hi-water mark dial attempts
   308 Max Dial fails before failover or Reboot
     0 Cumulative dial script fails
```

- Display additional device statistics

  Possible *stat* sub-commands:

```
app_security: Application layer
Security Statistics
app_sysvar: app_sysvar Statistics
beacon: Beacon Statistics
c1219: C1219 Meter Interface Statistics
c1219_timesync: C1219 Timesync
Statistics
comm_server: Comm Server Statistics
dhcp: DHCP Statistics
dli: DLI Statistics
dli_tr: DLI Traceroute Statistics
dll: FHSS DLL Statistics
dnp3: DNP3 Statistics
dns: DNS Statistics
esp: ESP net_mgr Statistics
esp_cost: ESP Cost Statistics
esp_mirror: ESP Mirroring Statistics
eth: Ethernet Statistics
gu: Gateway Serial Port Statistics
han_2_4: HAN 2.4 Statistics
han_2_4_ping: HAN_2_4 Ping Statistics
img_mgr: Image Management Statistics
img_rvt: Image Revert Statistics
imu_cpd: IMU CPD Statistics
ipv6: IPv6 Statistics
irq: IRQ Statistics
kio: KIO Statistics
ku: C1219 Serial Port Statistics
mac: FHSS MAC Statistics
mcp: Meter Comm Port Statistics
mlme: MLME Statistics
mlme_ext: MLME Extended Statistics
mlme_lg: MLME Last Gasp Statistics
mlme_sec: Link Layer Security
Statistics
mlme_sec_fsu: Maintenance Link Layer
Statistics
ndxp: Node Data Exchange Protocol
Statistics
net_bcast: Per-AP Broadcast Statistics
net_mgr: Net Manager Statistics
nexthop: L2 Nexthop Statistics
nic_meter: Nic Meter Statistics
nic_traffic: NIC Traffic Statistics
nid_log: NID Event Log Statistics
nm_trap: SSN Trap Statistics
ntp: SNTP Client update Statistics
phy: FHSS Physical Layer Statistics
rip: RIP Statistics
ru: Raw Serial Port Statistics
snmp_agent: SNMP agent Statistics
snmp_trap: SNMP trap Statistics
srt: SSN Routing Statistics
ssi: Serial Socket Statistics
timesync: Timesync Statistics
uart: Serial Port Driver Statistics
```


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Super User Cases

To execute these commands you need a PrivMgr certificate. For guidance on how to create one, please reference the CA Tools documentation.

- To close the clear text port
  - net_mgr -d *<ipv6 address>* conf nm_sec disable_unsecure 1
- To open the clear text port
  - net_mgr -d *<ipv6 address>* conf nm_sec disable_unsecure 0

Potential Errors and Troubleshooting

- *Response timed out*
  - Device did not respond to the NetManager command within the timeout period
  - The device may have intermittent reachability, or was in an idle state
    - Retry the command, optionally increase the timeout value with the *–t* flag
  - The device may not be reachable for various reasons: operational status, local conditions, maintenance, etc.
    - Check basic network connectivity (ping6, traceroute, etc)
    - Check device status in AMM
- *Erroneous Request*
  - This error may occur when running commands not intended for the type of device being queried (i.e. forcing ntp time sync on a non-AP device)
  - This error may be encountered when running commands on a device that are not supported by it's current firmware version (i.e. uploading signed blobs to an AP running < 2.10 FW)
    - Omit commands for signed blobs, if current firmware version is desired
    - If the AP is running below the desired firmware, upgrade the device and retry commands(s)
  - NetManager will also return this error when attempting to set a SHA value on an AP with routing disabled. Enable routing and restart (See Section 1), then retry commands.

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